

Gaussian mean and variance MLE

$$p(y_1, \dots, y_N | \mu, \sigma^2) = p(D | \mu, \sigma^2) = \prod p(y_n | \mu, \sigma^2) = \prod \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_n - \mu)^2}{2\sigma^2}}$$

Taking the log to get an easier cost function:

$$\begin{aligned} \log p(y_1, \dots, y_N | \mu, \sigma^2) &= \sum_{n=0}^N -\log(\sqrt{2\pi\sigma^2}) - \sum_{n=0}^N \left(\frac{(y_n - \mu)^2}{2\sigma^2} \right) \\ &= -\frac{N}{2} \log(2\pi\sigma^2) - \sum_{n=0}^N \left(\frac{(y_n - \mu)^2}{2\sigma^2} \right) \end{aligned}$$

Optimizing:

$$\hat{\mu} = \frac{\partial}{\partial \mu} \left\{ -\frac{N}{2} \log(2\pi\sigma^2) - \sum_{n=0}^N \left(\frac{(y_n - \mu)^2}{2\sigma^2} \right) \right\} = 0,$$

$$\frac{\partial}{\partial \mu} \left\{ -\sum_{n=0}^N \frac{(y_n - \mu)^2}{2\sigma^2} \right\} = -\frac{1}{2\sigma^2} \frac{\partial}{\partial \mu} \left\{ \sum_{n=0}^N (y_n - \mu)^2 \right\} = \frac{1}{\sigma^2} \sum_{n=0}^N (y_n - \mu) = 0$$

$$\boxed{\hat{\mu} = \frac{\sum_{n=0}^N y_n}{N}}$$

Plugging this result back into $p(D | \mu, \sigma^2) \Rightarrow p(D | \hat{\mu}, \sigma^2)$

$$\log p(D | \hat{\mu}, \sigma^2) = -\frac{N}{2} \log(2\pi\sigma^2) - \sum_{n=0}^N \left(\frac{(y_n - \hat{\mu})^2}{2\sigma^2} \right)$$

Optimizing:

$$\frac{\partial}{\partial \sigma^2} \left\{ \log p(D | \hat{\mu}, \sigma^2) \right\} = \underbrace{-\frac{\partial}{\partial \sigma^2} \left\{ \frac{N}{2} \log(2\pi\sigma^2) \right\}}_{N/2\sigma^2} - \underbrace{\frac{\partial}{\partial \sigma^2} \left\{ \sum_{n=0}^N \left(\frac{(y_n - \hat{\mu})^2}{2\sigma^2} \right) \right\}}_{+\frac{1}{2\sigma^4} \sum_{n=0}^N (y_n - \hat{\mu})^2} = 0$$

$$\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \cdot \sum_{n=0}^N (y_n - \hat{\mu})^2 = 0$$

$$\boxed{\hat{\sigma}^2 = \frac{\sum_{n=0}^N (y_n - \hat{\mu})^2}{N}}$$